3. Electrochemistry

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ECE-A
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① Electrochemistry is a branch of physical chemistry which deals with conversion of chemical energy to electrical energy and electrical energy to chemical energy.

Electric conductor :- The substance which allow electric current and pass through them are called "electric conductor."

These are of two thypes. They are :-
1, Metallic / electronic conductors.
2, Electrolytic conductors / electrolytes.

Electrodes :- Electrodes are solid substances. Generally metals acts as electrodes.

Electrodes are two types i, Anode ii, cathode

i, **Anode** :- The electrode at which oxidation takes place is called anode.

ii, **cathode** :- The electrode at which reduction takes place is called cathode.

Oxidation :- Removal of electrons is called oxidation.

Reduction :- Addition of electrons is called Reduction.

Redox reaction :- A reaction which involves simultaneous oxidation and Reduction is called Redox reaction.

Eg →

$$\overbrace{Zn + Cu^{2+} \rightleftharpoons Zn^{2+} + Cu}^{\text{oxi}}$$

$$\underbrace{Cu^{2+} \rightleftharpoons Cu}_{\text{Red}}$$

Electrolyte :- The substance which decomposes by the passage of electric current is called electrolyte.

Electrolysis :- The substance / phenomenon in which decomposition of electrolyte by the passage of electric current is called electrolysis.

Reference electrodes :-

The electrodes whose potential is arbitrarily taken as zero (or) Exactly known are called Reference electrodes.

Reference electrodes are two types.

(a) Primary Reference electrode

The electrode whose potential is arbitrarily taken as zero are called primary Reference electrode.

Eg - standard hydrogen electrode [SHE]

(b) Secondary Reference electrode

The electrode whose potential is not equal to zero but exactly known are called secondary Reference electrode.

Eg - calomel electrode
Ag / AgCl electrode
Glass electrode

* At first the potential of secondary reference electrode can be measured by using SHE after itself it act as secondary reference electrode.

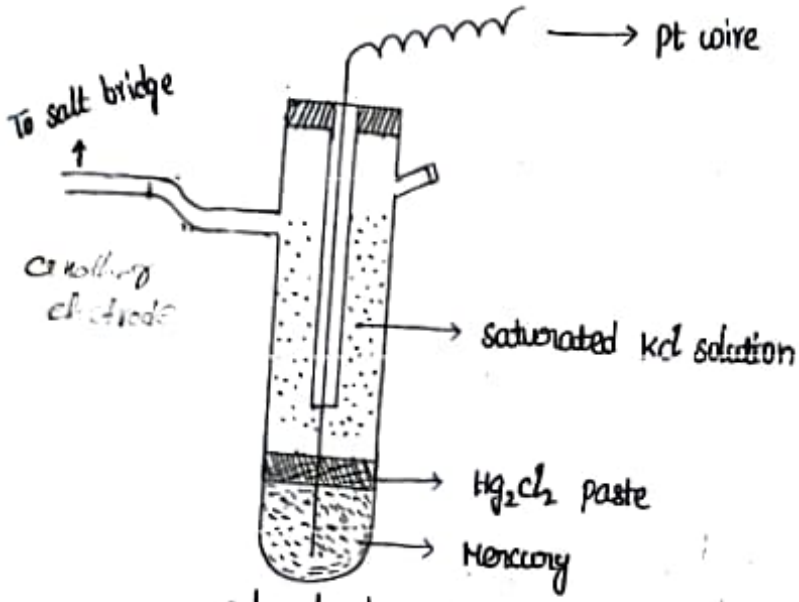
Calomel electrode

* It is a secondary reference electrode.
* Calomel electrode consists of long glass tube having side tubes on each side.
* The right side tube is used to fill the KCl solution.
* The left side tube is used to connect with other electrode through salt bridge whose potential is to be determined.
* The bottom of the glass tube, high purity mercury is placed. Above the mercury a paste of mercurious chloride [usually called calomel] is placed.
* The platinum wire sealed in glass tube is dipped into the mercury the outerpart of the pt wire is extended from the top used for electrical connection.

* The rest of the long glass tube is filled with a solution of KCl, acts as electrolyte.
* Top of the glass tube is closed.
* The calomel electrode is represented as

$$Pt / Hg_{(l)} / Hg_2Cl_{2\,(s)} / KCl_{(soln)}$$

* The electrode potential of calomel electrode is +0.2422V.



Calomel electrode

Working :-

1) Calomel electrode acts as both Anode and cathode depends on the connection with other electrode.
2) If it acts as Anode oxidation takes place

$$2Hg_{(l)} + 2Cl^- \rightleftharpoons Hg_2Cl_{2\,(s)} + 2e^-$$

3) If it acts as cathode Reduction takes place

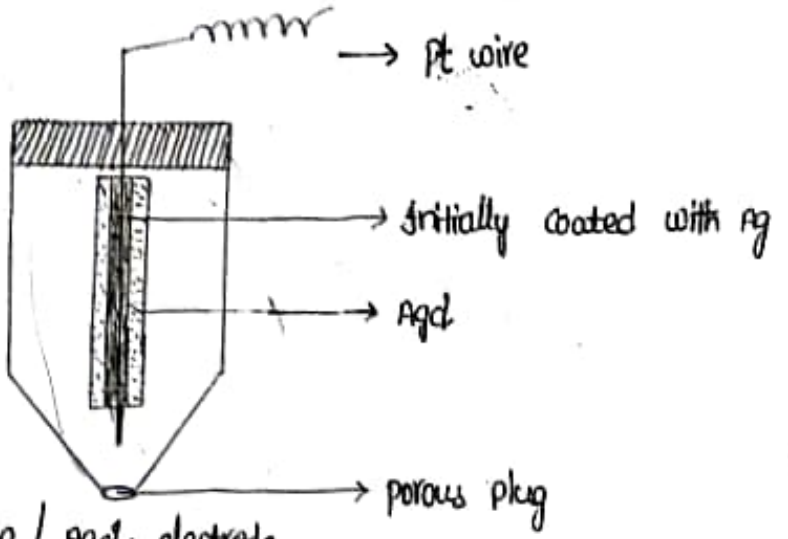
$$Hg_2Cl_{2\,(s)} + 2e^- \rightleftharpoons 2Hg_{(l)} + 2Cl^-$$

4) The potential of calomel electrode depends on the conc. of KCl solution.

For 0.1N KCl solution	E = +0.3338V
1.0N KCl solution	E = +0.2800 V
Saturated KCl	E = +0.2422 V

Ag / AgCl electrode :- | silver / silver chloride

* Ag / AgCl electrode is secondary reference electrode.
* It is also called as metal / metal salt ion electrode.
* It contains hard glass tube and the bottom of the glass tube porous plug is present used to connect another electrode whose potential is to be determined.
* A platinum wire is initially coated with Ag and again coated with AgCl solution is placed in glass tube.
* The outer part of platinum wire is extended from the top of the glass tube, used to connect electrical contact.
* The rest of the glass tube is filled with KCl solution.
* The top of the glass tube is covered.
* It is represented as Ag / AgCl / KCl (or) HCl



Ag / AgCl electrode

Working :-

1) Ag / AgCl electrode acts as both anode and cathode depends on the connection with other electrode.

2) If it acts as anode oxidation takes place

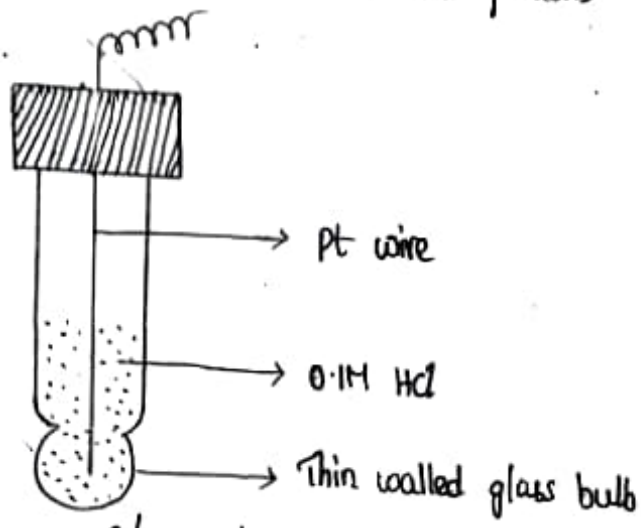
$$Ag^+ + Cl^- \longrightarrow AgCl + e^-$$

3) If it acts as cathode reduction takes place

$$AgCl + e^- \longrightarrow Ag^+ + Cl^-$$

Glass electrode :-

* A glass electrode is specially used for determining pH value of the solution. so it act as internal Reference electrode.
* The glass electrode consists of a thin walled glass bulb made up of a special kind of low melting point glass.
* The glass bulb is filled with 0.1M HCl solution.
* The platinum wire dipped in the electrolyte passes out of the glass tube.
* It is represented as Pt, 0.1M HCl / Glass



Glass electrode

(2) Electrochemical cell :-

An electrochemical cell is a device in which chemical energy is converted into Electrical energy through redox reactions.

Eg → Galvonic cell (or) voltaic cell

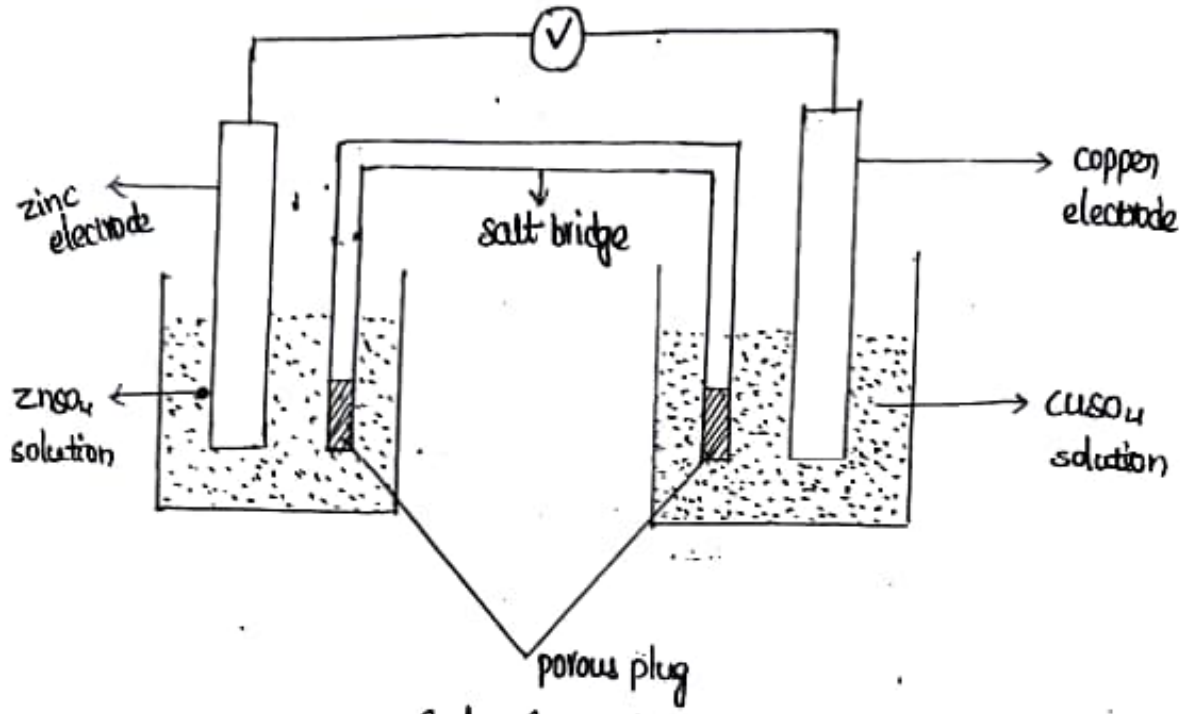
Galvonic cell (or) voltaic cell :-

A cell (or) device which converts chemical energy to electrical energy through redox reactions is called Galvonic (or) voltaic cell.

* It is a type of electrochemical cell.

Construction

* Galvonic cell consists of two half cells.
* one Half cell consist 'Zn' electrode dipped in $ZnSO_4$ solution.
* Another Half cell consist copper electrode dipped in $CuSO_4$ solution
* $ZnSO_4$ and $CuSO_4$ solutions acts as an electrolytes.
* The two electrodes are connected by a wire through voltmeter and the two solutions are connected by a salt bridge.



Galvonic cell

Working :-

At Anode

Oxidation takes place in the zinc electrode by the liberation of electrons, so this electrode is called negative electrode. This is called oxidation half cell reaction.

$$Zn \longrightarrow Zn^{2+} + 2e^- \longrightarrow \text{oxidation}$$

to loss of electrons

At cathode

Reduction takes place in the copper electrode by the gaining of electrons which was lost by the Zn electrode, so this electrode is called positive electrode. This is called Reduction half cell reaction

$$Cu^{2+} + 2e^- \longrightarrow Cu \longrightarrow \text{Reduction}$$

At anode : $Zn \longrightarrow Zn^{2+} + 2e^-$

At cathode : $Cu^{2+} + 2e^- \longrightarrow Cu$

net cell reaction : $Zn + Cu^{2+} \longrightarrow Zn^{2+} + Cu$

The electrons liberated at anode by zinc electrode flow through the external wire and are consumed by the copper electrode at the cathode.

The flow of electrons from one electrode to another electrode causes the generation of electric current (or) electricity.

representation of Galvanic cell

$$Zn_{(s)} \,/\, ZnSO_4{}_{(1M)} \,//\, CuSO_4{}_{(1M)} \,/\, Cu_{(s)}$$

(or)

$$Zn \,/\, Zn^{2+} \,//\, Cu^{2+} \,/\, Cu$$

cell potential (or) EMF of cell :-

The difference of potential which causes flow of electrons from higher potential electrodes to lower potential electrode is called the EMF of the cell (or) cell potential.

EMF means electromotive force.

It is represented as E°_{cell}.

EMF = standard reduction potential of right hand side electrode — standard reduction potential of left hand side electrode

(or)

$$E^\circ_{cell} = E^\circ_{right} - E^\circ_{left}$$

(or)

$$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{Anode}$$

Nernst equation

The equation which gives the relation b/w electrode potential and concentration of electrolyte is called Nernst Equation.

$$E_{cell} = E^{\circ}_{cell} - \frac{2.303RT}{nF} \log Q \text{ (or) } C$$

(or)

$$E_{cell} = E^{\circ}_{cell} - \frac{0.0591}{n} \log Q \text{ (or) } \log C$$

where,

E_{cell} = cell potential

E°_{cell} = standard cell potential

R = Gas constant [8.314 J/mol]

n = no. of e^-'s involved in reaction

F = Faraday [1F = 96,500C]

Q = Reaction Quotient (or) $\frac{[Product]}{[Reactant]}$

Nernst Equation :-

$$E_{cell} = E^{\circ}_{cell} - \frac{0.0591}{n} \log \frac{[P]}{[R]}$$

E_{cell} → Emf Cell

E°_{cell} → Standard Emf Cell

n → no of e^- Transfers reaction.

Derivation :-

$$aA + bB \rightleftharpoons cC + dD$$

$$Q = \frac{[\text{product}]}{[\text{Reactants}]} = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

Thermodynamic Eq'n

$$\Delta G = \Delta G^{\circ} + RT \ln Q \longrightarrow (1)$$

R → Gas constant

T → Temperature.

ΔG → Change in Gibbs free energy

ΔG° → Change in Standard Gibb's free energy.

$$\Delta G = -nFE \rightarrow (2)$$

$$\Delta G^{\circ} = -nFE^{\circ} \rightarrow (3)$$

eq ② & ③ in ①

$$-nFE_{cell} = -nFE^{\circ}_{cell} + RT\ln Q \rightarrow (4)$$

Dividing eq ④ "$-nF$"

$$\frac{-nFE_{cell}}{-nF} = \frac{-nFE^{\circ}_{cell}}{-nF} - \frac{RT}{nF}\ln Q$$

$$E_{cell} = E^{\circ}_{cell} - \frac{RT}{nF}\ln\frac{[P]}{[R]}$$

$$E_{cell} = E^{\circ}_{cell} - \frac{2.303RT}{nF}\log\frac{[P]}{[R]}$$

at 25°c.

$$E_{cell} = E^{\circ}_{cell} - \frac{0.0591}{n}\log\frac{[P]}{[R]}$$

Single Electrode :-

(Anode half cell) → Oxidation.

$$M_{(s)} \longrightarrow M^{+n}_{(aq)} + ne^-$$

Apply Nearnest eq'n

$$E_{cell} = E^{\circ}_{cell} - \frac{0.0591}{n}\log\frac{[M^{+n}]}{[M]}$$

No of Solids Gases (M) = 1

$$E_{cell} = E^{\circ}_{cell} - \frac{0.0591}{n}\log\frac{[M^{+n}]}{1}$$

$$E_{cell} = E^{\circ}_{cell} - \frac{0.0591}{n}\log[M^{+n}]$$

(cathode half cell) → Reduction

$$M^{+n} + ne^- \rightleftharpoons M$$

Apply Nearnest eq'n

$$E_{cell} = E^{\circ}_{cell} - \frac{0.0591}{n}\log\frac{[M]}{[M^{+n}]}$$

$$E_{cell} = E^{\circ}_{cell} - \frac{0.0591}{n}\log\frac{1}{[M^{+n}]}$$

$$E_{cell} = E^{\circ}_{cell} + \frac{0.0591}{n}\log[M^{+n}]$$

$$E_{Oxidation} = E^{\circ}_{Oxd} - \frac{0.0591}{n} \log [Zn^{+2}]$$

$$E_{Red} = E^{\circ}_{Red} + \frac{0.0591}{n} \log [Cu^{+2}]$$

$$E_{cell} = E_{Oxi} + E_{Red}$$

$$E_{cell} = E^{\circ}_{Oxi} - \frac{0.0591}{n} \log [Zn^{+2}] + E^{\circ}_{Red} + \frac{0.0591}{n} \log [Cu^{+2}]$$

$$E_{cell} = E^{\circ}_{Oxi} + E_{Red} - \frac{0.0591}{n} \log [Zn^{+2}] + \frac{0.0591}{n} \log [Cu^{+2}]$$

$$E_{cell} = E^{\circ}_{cell} - \frac{0.0591}{n} \log [Zn^{+2}] + \frac{0.0591}{n} \log [Cu^{+2}]$$

$$E_{cell} = E^{\circ}_{cell} - \frac{0.0591}{n} \left[\log [Zn^{+2}] - \log [Cu^{+2}]\right]$$

$$\boxed{E_{cell} = E^{\circ}_{cell} - \frac{0.0591}{n} \log \frac{[Zn^{+2}]}{[Cu^{+2}]}}$$

ive electrode

Single electrode potential :-

The tendency (or) capacity of an electrode to lose or gain electrons, when it is in contact with ocon ion solution is called single electrode potential.

Standard electrode potential :-

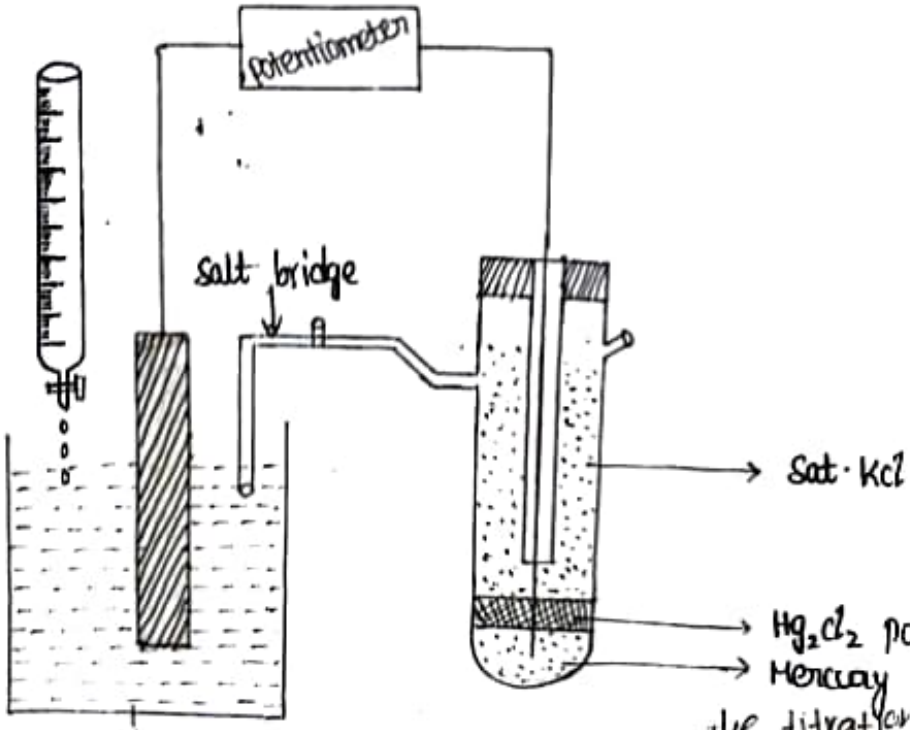
The tendency (or) capacity of an electrode to lose or gain electrons when it is in contact with ocon ion solution at standard conditions like 1 molar concentration and temp 25°c is called standard electrode potential.

Potentiometric titrations :-

The titrations in which end point is determined based on the potential difference b/w two electrodes with the addition of the titrant are called Potentiometric titrations.

Eg → Redox titration

Redox titration [$FeSO_4$ vs $K_2Cr_2O_7$]



Potentiometer titration

the titration in which the end points can be determined the potential value of the solution

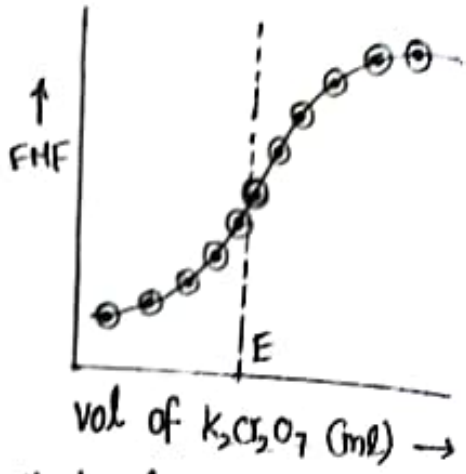
Ex: redox titration

* potentiometric titration [$FeSO_4$ vs $K_2Cr_2O_7$] known amount of $FeSO_4$ solution is taken in a beaker.
* To this an indicator electrode [platinum electrode] is dipped in it.
* It is then connected to a reference electrode [calomel electrode] to form a Galvonic cell. Both are connected to the potentiometer and its potential is determined.
* when this solution is titrated against the standard $K_2Cr_2O_7$ solution taken in the burette the emf is going on increasing as the conc. of Fe^{3+} increases.

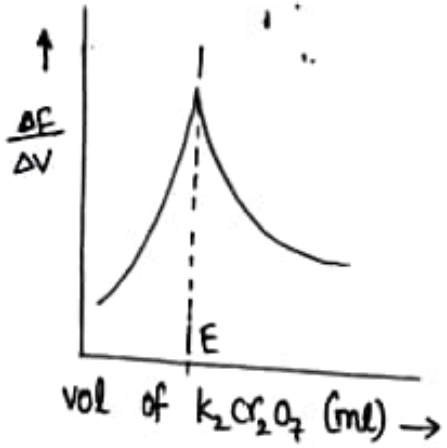
$$Fe^{2+} \xrightarrow{K_2Cr_2O_7} Fe^{3+} + e^-$$

* At the end point the emf is suddenly increases. After the end point there is no change in the potential.

* When the emf is plotted against the volume of $K_2Cr_2O_7$ added a curve obtained as shown in the fig. The end point is the point where the ~~shape~~ slope of the curve is maximum.



* A more sensitive and satisfactory method of detecting the end point will be the given graph of $\frac{\Delta E}{\Delta V}$ against volume of $K_2Cr_2O_7$. The resulting curve raises a maximum at the equivalent point which is the end point.



Concept of conductivity :-

The conductivity is an indicator of how easily current passes through the solution.

* The conductivity mainly depends on the quantity of ions and their mobility in that solution.

conductance [C] :-

→ The reciprocal of resistance is called conductance.

→ It is represented as "C".

$$C = \frac{1}{R}$$

Units ⇒ ohm^{-1} (or) mho

Conductometric titration :-

2/5 The titration in which end point is determined based on the measurement of conductance of the solution during the titration is called conductometric titrations.

→ It is a volumetric method.

Principle :- The conductance of solution depends on
(i) The number and charge on the free ions and
(ii) The mobility of the ions.

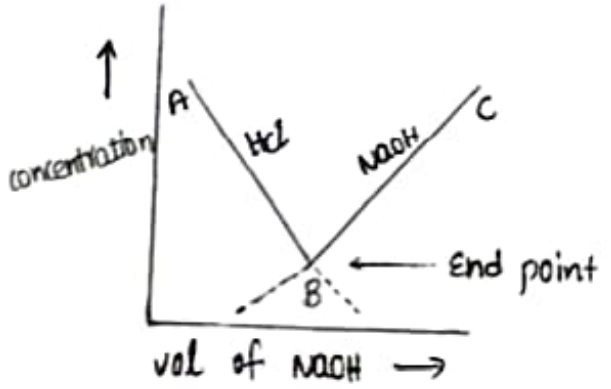
Acid - Base titrations :-

(i) strong acid Vs strong base [HCl Vs NaOH]

* Known amount of acid [HCl] is taken in the conductivity cell and base [NaOH] in the burette.
* Initially the conductivity of HCl is high, this is due to the fast moving H^+ ions [point 'A' in the graph].
* As the NaOH added gradually, conductance will be going on decreasing until the acid has been completely neutralised [Indicated by the line AB].
* This is due to the replacement of fast moving H^+ ions by slow moving Na^+ ions.
  The point 'B' indicates complete neutralisation of all H^+ ions.

$$\underset{[Acid]}{HCl} + \underset{[Base]}{NaOH} \longrightarrow \underset{[salt]}{NaCl} + \underset{[water]}{H_2O}$$

* Further addition of NaOH will introduce the fast moving OH^- ions. Therefore the conductance will be going on increasing [line AB].
* On plotting the graph b/w conductance and the volume of NaOH added, the two lines intersect at a point B gives the end point.
* This point gives the volume of NaOH required for neutralisation.



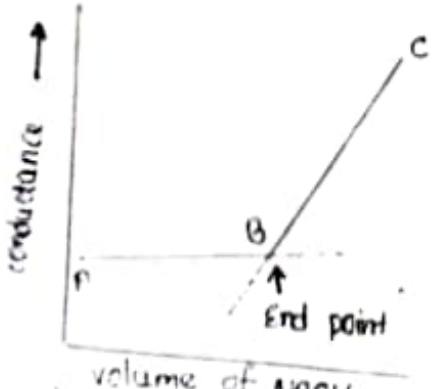
Strong acid Vs strong Base

(2) weak acid Vs strong base [CH_3COOH Vs NaOH]

* known amount of weak acid [CH_3COOH] is taken in the conductivity cell and base [NaOH] in the burette.
* Initially, the conductivity of CH_3COOH is low, this is due to poor dissociation of CH_3COOH. [point 'A' in graph].
* As NaOH is added gradually, conductance will be going on slow increasing [indicated by the line AB]. This is due to the formation of highly ionised CH_3COONa.
* The point 'B' indicates complete neutralisation of CH_3COOH

$$\underset{\text{weak acid}}{CH_3COOH} + \underset{(s.b)}{NaOH} \longrightarrow \underset{\text{(highly ionised salt)}}{CH_3COONa} + H_2O$$

* Further addition of NaOH introduces excess of fast moving OH^- Therefore the conductance sharply increases. [line BC].
* on plotting the graph b/w conductance against the volume of NaOH added the two lines intersect at point 'B' gives the end point.



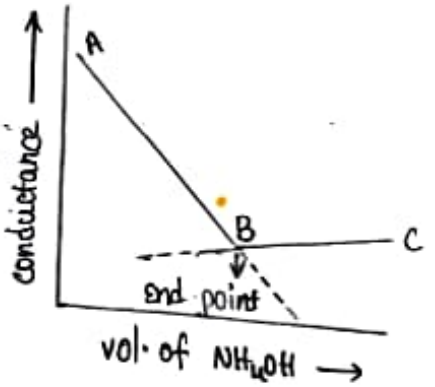
W.a Vs S.b

Strong acid Vs weak base [HCl Vs NH_4OH]

* known amount of HCl is taken in the conductivity cell and NH_4OH in the burette.
* Initially the conductivity of HCl is high, this is due to the fast moving H^+ ions [point 'A' in the graph]
* As NaOH added gradually conductance will be going on decreasing until the acid has been completely neutralised [line AB]
* This is due to the replacement of fast moving H^+ ions by slow moving NH_4^+ ions.
* The point 'B' indicates complete neutralisation of all H^+ ions.

$$\underset{(S.A)}{HCl} + \underset{(w.b)}{NH_4OH} \longrightarrow \underset{(salt)}{NH_4Cl} + H_2O$$

* Further addition of weak NH_4OH does not effect change in the conductance [line BC]
* on plotting the graph b/w conductance and the volume of NaOH added, the two lines intersect at a point 'B' gives the end point.

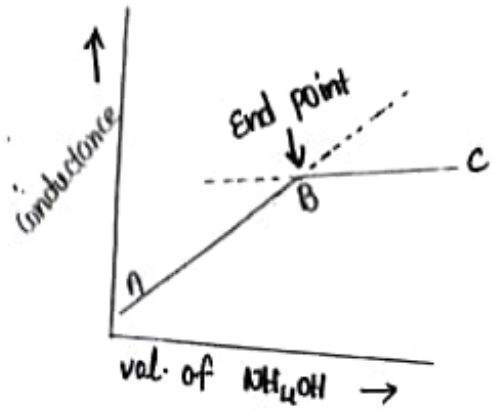


S.A Vs w.b

weak acid Vs weak base [CH_3COOH Vs NH_4OH]

* known amount of CH_3COOH is taken in the conductivity cell and NH_4OH in the burette.
* Initially the conductivity of CH_3COOH is low, this is due to the poor dissociation of CH_3COOH [point 'A' in graph]

* As ammonia added gradually conductance will be going on slowly [indicated by the line AB]. This is due to the formation of ionised salt CH_3COONH_4.
* After the neutralisation of CH_3COOH, further addition of NH_4OH does not affect any change in the conductance [line BC].
* On plotting the graph b/w conductance against the volume of NH_4OH added, the two lines intersect at point 'B' gives the end point.



W.A Vs W.b

$$\underset{(W.a)}{CH_3COOH} + \underset{(W.b)}{NH_4OH} \longrightarrow \underset{\text{(Highly ionised salt)}}{CH_3COONH_4} + H_2O$$

Batteries

Definition :-

A device which construct with single electrochemical cell or several electrochemical cells are inter connected in a series through electrodes is called Battery.

Eg - Pb-acid battery, Zn - Air battery

Types of Batteries

Batteries are classified into two types.

They are :- 1) Primary battery
2) Secondary battery

Primary Battery

The cells in which the electrode reactions are cannot be reversed by passing external energy, are called primary battery.

* In these cells the reactions occur only once and after use they become dead.
* Hence these are also called not rechargable batteries.

Eg → Dry cell, zinc air battery etc.

2) Secondary Battery

The cells in which the electrode reactions are reversed by passing external energy are called secondary battery.

* In these cells the reaction occur in forward as well as backward.
* Hence these are recharged by passing electric current and used again and again.

Eg → Pb - acid storage battery, Ni-Cd battery

Zinc - air battery

Zinc - air battery work by the oxidation of Zn with oxygen from the air.

Construction :-

1) In this cell 'Zn' acts as anode.
2) porous carbon rod acts as cathode which access oxygen from the air.
3) KOH acts as an electrolyte.

Working :-

At Anode :- oxidation takes place at anode. Zn undergo oxidation, releasing electrons to travel to the cathode

$$Zn \rightarrow Zn^{2+} + 2e^-$$

$$Zn^{2+} + 2OH^- \rightarrow ZnO + H_2O \quad \text{(Oxidation)}$$

$$\text{(Reduction)}$$

$$Zn + 2OH^- \rightarrow ZnO + H_2O + 2e^-$$

reacts at the cathode and form OH^- ions.

$$\frac{1}{2} O_2 + H_2O + 2e^- \longrightarrow 2OH^- \quad \text{(Reduction)}$$

Net cell reaction :

$$Zn + 2OH^- \longrightarrow ZnO + H_2O + 2e^- \quad \text{[Oxidation]}$$
$$\frac{1}{2} O_2 + H_2O + 2e^- \longrightarrow 2OH^- \quad \text{[Reduction]}$$

$$Zn + \frac{1}{2} O_2 \longrightarrow ZnO$$

Discharging voltage is 1.65 V

Examples for secondary Battery

Nickel - cadmium Battery → It is a secondary battery.

Construction :-

1) Ni-cd battery consist two electrodes namely anode and cathode and an electrolyte.
2) Cadmium acts as anode and $NiO(OH)_2$ [Nickel oxide hydroxide] acts as cathode.
3) Metal salt (or) KOH acts as an electrolyte.

Working

At Anode :

At cathode :

$$Cd + 2OH^- \longrightarrow Cd(OH)_2 \quad \text{(Oxidation)}$$
$$NiO(OH)_2 + 2H_2O \longrightarrow Ni(OH)_2 + 2e^- + 2OH^- \quad \text{(Reduction)}$$

Net cell reaction :- $Cd + NiO(OH)_2 + 2H_2O \longrightarrow Ni(OH)_2 + Cd(OH)_2 + 2e^-$

Lithium ion batteries :

It is a secondary battery.

In lithium battery it does not contain metallic lithium as anode, but the movement of lithium ions are responsible for charging and discharging.

Construction

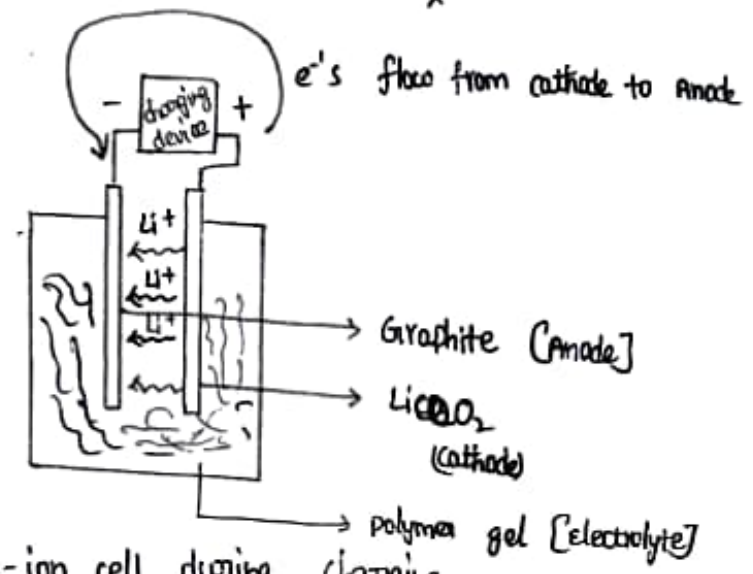
1) Li-ion Battery consist two electrodes namely Anode and cathode, and an electrolyte.
2) The layers of porous carbon [graphite] acts as anode.
3) Layers of Lithium metal oxide [Lithium cobalt monoxide] acts as cathode.
4) polymer gel acts as an electrolyte.

Working

Charging

* During charging Li^+ ions flow from the cathode [$LiCoO_2$] to anode [graphite].
* The electrons also move from cathode to Anode.
* The electrons and Li^+ ions combine at the Anode and deposit there as Li.

$$LiCoO_2 + C \longrightarrow Li_{1-x}CoO_2 + \underset{\text{graphite [anode]}}{CLi_x}$$

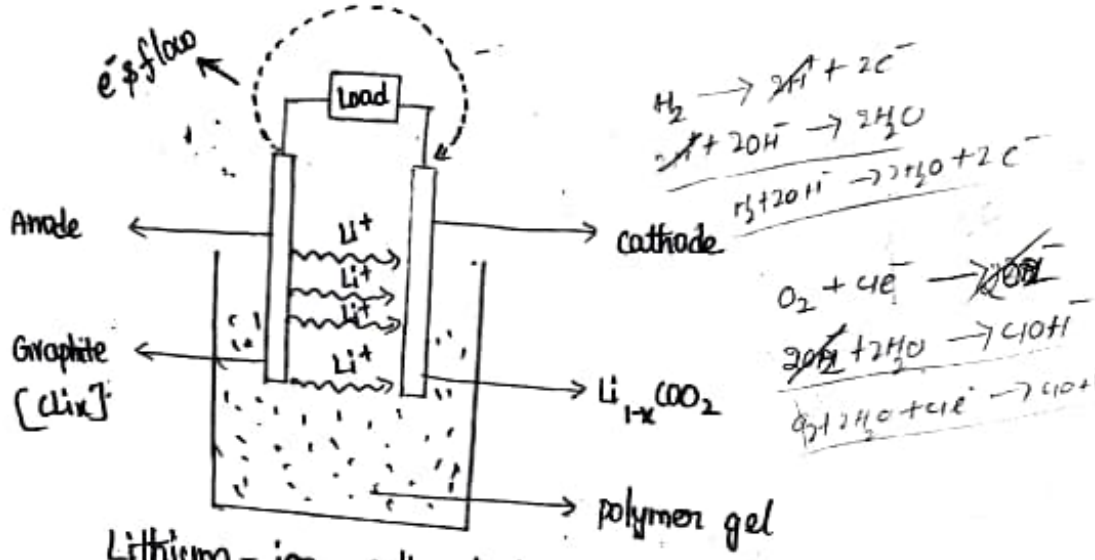


Lithium-ion cell during charging

Discharging

* During discharging Li^+ ions flow back through the electrolyte from Anode to cathode.
* The electrons as flow from Anode to cathode.
* The electrons and Li^+ ions combine at the cathode & depo as Li.

$$Li_{1-x}CoO_2 + CLi_x \longrightarrow LiCoO_2 + C$$



Lithium - ion cell during discharging

Fuel cells (or) fuel Battery :- [long time energy sources]

1) A device which convert chemical energy directly into electrical energy without combustion is called fuel cell.
2) Large number of fuel cells are connected in a series to form fuel battery.

Fuel + oxygen ⟶ oxidation product + electricity.

Eg → H_2-O_2 fuel cells

CH_3OH - O_2 fuel cells

1) Hydrogen - oxygen fuel cell (or) H_2-O_2 fuel cell

Construction :- This cell consist two porous electrodes namely anode & cathode & an electrolyte.

* high weight fuel battery
* cost is very high

The porous carbon rods coating with small amount of catalyst (10)
[Pt, Pd, Ag] acts as Anode & cathode.
25% K_2OH (or) NaOH acts as electrolyte.
These two electrodes are connected through volt meter.

Working

At Anode :- oxidation takesplace at anode, when H_2 gas [fuel] is passed through the anode is oxidised with the libration of electrons.

→ It is then combine with OH^- ions to form water.

$$H_2 \longrightarrow 2H^+ + 2e^-$$
$$2H^+ + 2OH^- \longrightarrow 2H_2O$$
$$\overline{H_2 + 2OH^- \longrightarrow 2H_2O + 2e^-}$$

$H_2 \rightarrow 2H^+ + 2e^-$
$2H^+ + 2OH^- \rightarrow 2H_2O$

$O_2 + 4e^- \rightarrow 2O_2^-$
$2O_2^- + 2H_2O \rightarrow 4OH^-$

$O_2 + 4e^- \rightarrow 2O_2^-$
$2O_2^- + 2H_2O \rightarrow 4OH^-$

The above r/c is multiplied by "2"

$$2H_2 + 4OH^- \longrightarrow 4H_2O + 4e^- \quad \text{(At Anode)}$$

At cathode :- Reduction takes place at cathode. The Oxygen gas is passed through the cathode accepts electrons, which was obtained from Anode and reacts with water to form OH^- ions.

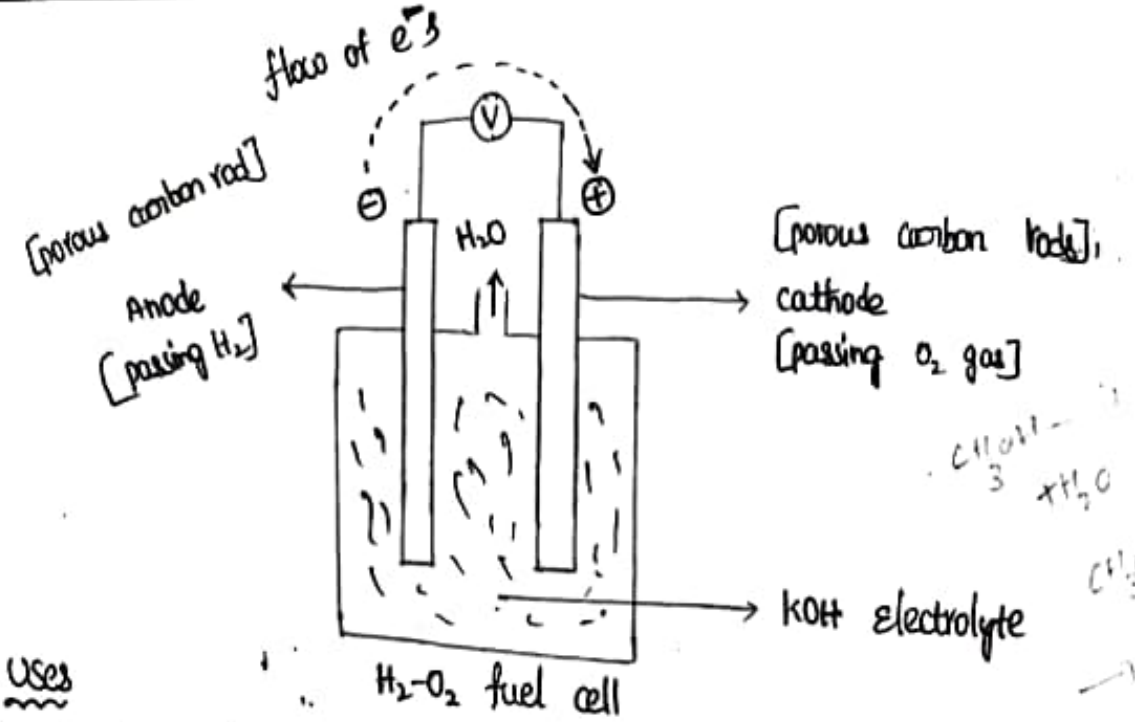
$$O_2 + 4e^- \longrightarrow 2O^{2-}$$
$$2O^{2-} + 2H_2O \longrightarrow 4OH^-$$
$$\overline{O_2 + 2H_2O + 4e^- \longrightarrow 4OH^-} \quad \text{(At cathode)}$$

$H_2 \rightarrow 2H^+ + 2e^-$
$2H^+ + 2OH^- \rightarrow 2H_2O$
$H_2 + 2OH^- \rightarrow 2H_2O + 2e^-$

Net cell r/c is :-

$$2H_2 + 4OH^- \longrightarrow 4H_2O + 4e^- \quad \text{(At Anode)}$$
$$O_2 + 2H_2O + 4e^- \longrightarrow 4OH^- \quad \text{(At cathode)}$$
$$\overline{2H_2 + O_2 \longrightarrow 2H_2O}$$

→ Discharging voltage is 0.8 to 1.0V



H_2-O_2 fuel cell

Uses

1) H_2-O_2 fuel cells are used as energy source in space vehicles, Submarines & military vehicles.

2) In H_2-O_2 fuel cells the water obtained is used as source of fresh water to the astronauts.

Methanol - oxygen fuel cell [or] (CH_3OH - O_2 fuel cell)

Construction :- This cell consist of two electrodes namely anode and cathode & an electrolyte

* The electrode consist of porous nickel plate coated with a catalyst Pt acts as anode.
* The other electrode consist of porous nickel plate coated with a catalyst Ag acts as cathode.
* 25% of NaOH (or) KOH acts as an electrolyte.

Working

At Anode :- CH_3OH and H_2O mixture is passed through the anode. oxidation takes place at with the liberation of electrons.

$$CH_3OH + H_2O \longrightarrow CO_2 \uparrow + 6H^+ + 6e^-$$

Cathode :- O_2 (or) air is passed through the cathode, reduction (R) takes place by gaining of electrons. These electrons are move from anode to cathode through an external circuit.

$$\frac{3}{2}O_2 + 6H^+ + 6e^- \longrightarrow 3H_2O$$

Net cell r/cs :

At Anode :

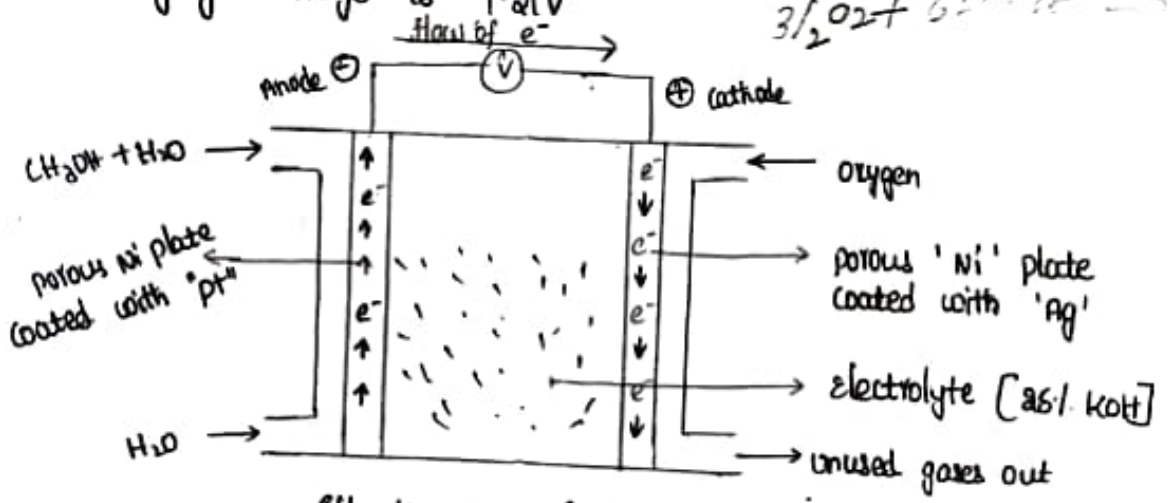
$$CH_3OH + H_2O \longrightarrow CO_2\uparrow + 6H^+ + 6e^- \quad \text{(oxidation)}$$

At cathode :

$$\frac{3}{2}O_2 + 6H^+ + 6e^- \longrightarrow 3H_2O \quad \text{(Reduction)}$$

$$CH_3OH + \frac{3}{2}O_2 \longrightarrow CO_2\uparrow + 2H_2O\uparrow$$

→ Discharging voltage is 1.21V



$CH_3OH - O_2$ fuel cell)

Sensors (or) Electrochemical sensors :-

Imp

A device which is used to defect (or) Analyse minute changes in physical and chemical properties of sample by electrical signals is called Electrochemical sensors.

Electrochemical sensors are mainly classified into two types.

(a) potentiometric sensors [If we measure difference of two potentials in volts]

(b) Amperometric sensors [If we measure the current in amperes]

1) Potentiometric sensors

* A sensor used to determine the analytical conc. of some compo in the solution is called potentiometric sensors.

1) potentiometric sensors also consist electrodes and an electr
2) One electrode is an ion selective electrode [ISE] and other electrode is reference electrode.
3) The ion selective electrode is used to detect the ions present in electrolyte usually H^+ ions.
4) The ISE is also called as membrane type electrode, gene glass membrane electrode is used as ISE.
5) The ISE is coupled with reference electrode to complete electrical circuit.
6) This sensor measured the potential difference b/w two electro
7) The potential arised b/w the electrodes is based on the concentration of H^+ ions present in electrolyte.
8) The potential difference b/w two electrodes produce electrical signals.

Eg :- Glucose sensor is example of potentiometer sensor.

Glucose sensor :-

* This sensor is used to detect the glucose.
* Glucose is an electrically neutral molecule. It is difficult to produce electrical signals.
* First these neutral molecules is converted into ions, these ions are selectively detected by the electrodes.
* It involves the following steps.

Step - 1

Enzyonatic oxidation of glucose.

Glucose react with oxygen in presence of glucose oxidase enzyme to form gluconic acid and H_2O_2 glucose oxidase.

$$D\text{-glucose} + O_2 + H_2O \longrightarrow D\text{-gluconic acid} + H_2O_2$$

Step - 2

Further decomposition of H_2O_2 gives H^+ ions.

$$H_2O_2 \longrightarrow 2H^+ + O_2 + 2e^-$$

Amperometric sensors

1) This sensor is based on the measurement of current b/w working electrodes and counter electrode.
2) In Amperometric sensors the signal is obtained by diffusion of current which is depends on the concentration of ions travelled electrochemically.
3) Eg → The example of amperometric sensor is detection of uric acid.

Estimation of uric acid

→ Uric acid is nitrogeneous compound in urine, which is related to many clinical disorders.

→ one of the major problem in the detection of uric acid is interference of ascorbic acid, which has similar oxidation potential.

→ These are two methods for determination of uric acid
(a) Enzymatic method
(b) Non - Enzymatic method

(a) Enzymatic method :- Uric acid can be Estimated in the presence of Enzyme uricase.

$$\text{Uric acid} + O_2 + H_2O \xrightarrow{\text{uricase}} \text{Allantoin} + H_2O_2 + CO_2$$

$$H_2O_2 \longrightarrow 2H^+ + O_2 + 2e^-$$

(b) Non - Enzymatic method :- In this method uric acid can be estimated by using differential pulse voltammetry.

* Because the differential pulse voltammetry peak currents of uric acid & ascorbic acid could be separated in this method.

Uses of E.C. sensors

1) Detection of oxygen and toxic gases like nitrogen oxides, sulphur oxides, carbon oxides & Hydrogen sulphide.
2) These are used to detect various biological human fluids.
3) These are also used to detect trace level concentration of various ingredients in fertilizers, pesticides & drugs.
4) These are used to detect glucose in blood & uric acid in urine.
5) These are used to detect climate changes in environment.
6) These are also used in agricultural industries etc.

$H_2 \longrightarrow 2H^+ + 2e^-$

$2H^+ + 2OH^- \longrightarrow 2H_2O$

$H_2 + 2OH^- \longrightarrow 2H_2O + 2e^-$

$O_2 + 4e^- \longrightarrow 2O^-$

$2O^- + 2H_2O \longrightarrow 4OH^-$

$O_2 + 4e^- + 2H_2O \longrightarrow 4OH^-$

$2H_2 + O_2 \longrightarrow 2H_2O$

1. minimal sound pollution
2. minimal visual "
3. charging takes little time
4. ability to adapt
5. reduction of power supply
6. extensive utilisation times

Fuell Cell

Polymer Electrolyte membarane Electrolytic Cell:

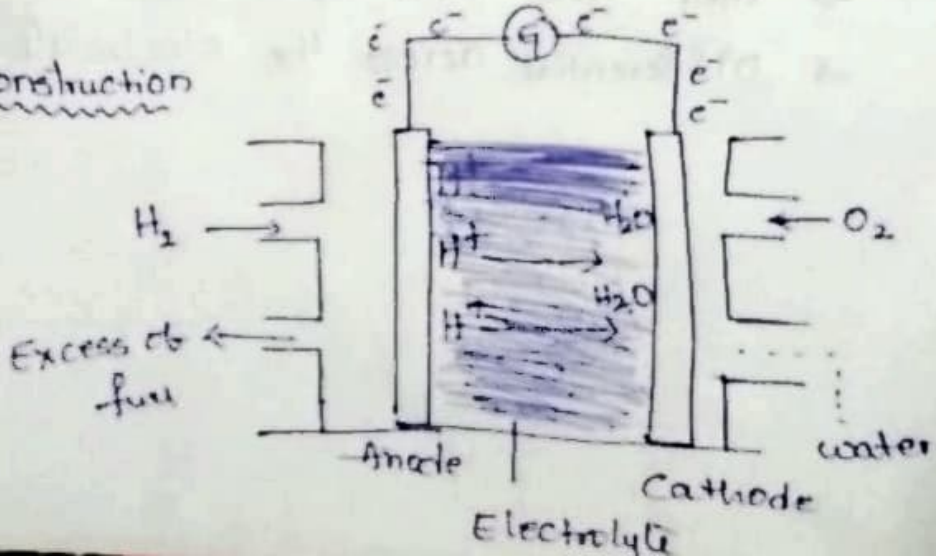
→ The protons (H^+ ions) move through Electrolytic to the Cathode to Combine with oxygen and Electrons to produce water

→ It is uses a polymeric membarane as the Electrolyte, with platinum Electrodes

→ These cells operates at relatively low temparature which allows them to start quickly and results in Less water on system Components, resulting best durability

→ It uses water-based acidic polymer membarane as it is Electrolyte, with platinum - based electrodes. The protons passed through the ~~mem~~ membarane at Cathode side of the Cell while the electrons travel in an external Circuit, generate electrical Current

Construction



Cell reactions :-

Anode reaction :-

$$H_2 \longrightarrow 2H^+ + 2e^-$$

Cathode reaction

$$\frac{1}{2} O_2 + 2H^+ + 2e^- \longrightarrow H_2O$$

overall Cell reaction

at anode :- $H_2 \longrightarrow \cancel{2H^+} + \cancel{2e^-}$

at Cathode : $\frac{1}{2} O_2 + \cancel{2H^+} + \cancel{2e^-} \longrightarrow H_2O$

$$H_2 + \frac{1}{2} O_2 \longrightarrow H_2O$$

$$2H_2 + O_2 \longrightarrow 2H_2O$$

Advantages :-

→ They have short response of time

→ They have higher purity of hydrogen

→ They have lower operational cost

→ Differential across the Electrolyte